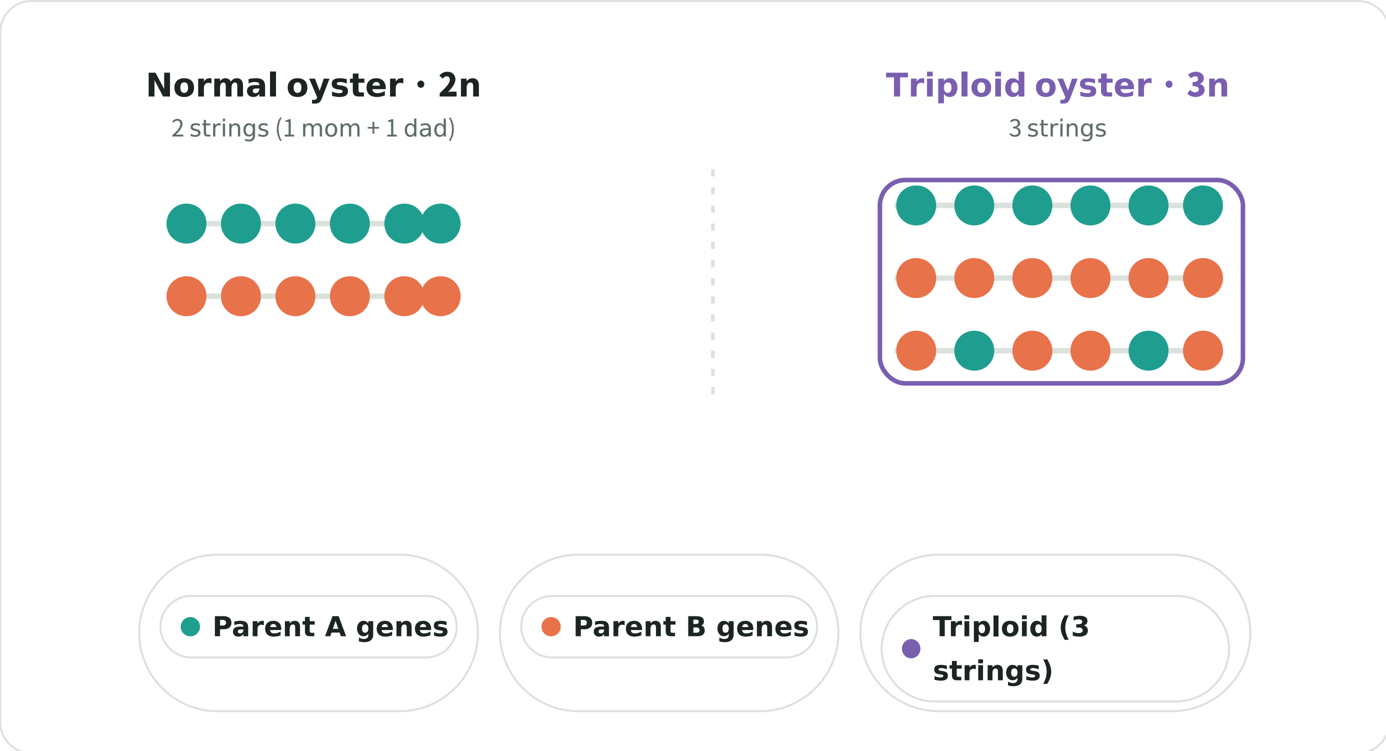
Is this triploid oyster really 3n?

Think of chromosomes as **strings of colored beads**. A normal oyster has 2 strings; a triploid has 3.



Two questions this tool answers

Question 1

Is it really 3 strings?

Check whether it is 2 strings (2n) or 3 (3n).

Question 2

Which parents, how much?

Count how many of the 3 strings come from A vs B.

The whole flow

Read DNA pieces → stick them on a map → answer the two questions.

1. **Read DNA**
   Sequence lots of short DNA pieces (reads).

2. **Stick on a map**
   Align pieces to a reference genome (mapping).

3. **Q1 · Triploidy check**
   Look at the mix ratio of two letters at one spot (nQuire).

4. **Q2 · Count parents**
   Use parent-only markers (diagnostic SNPs) to count A.

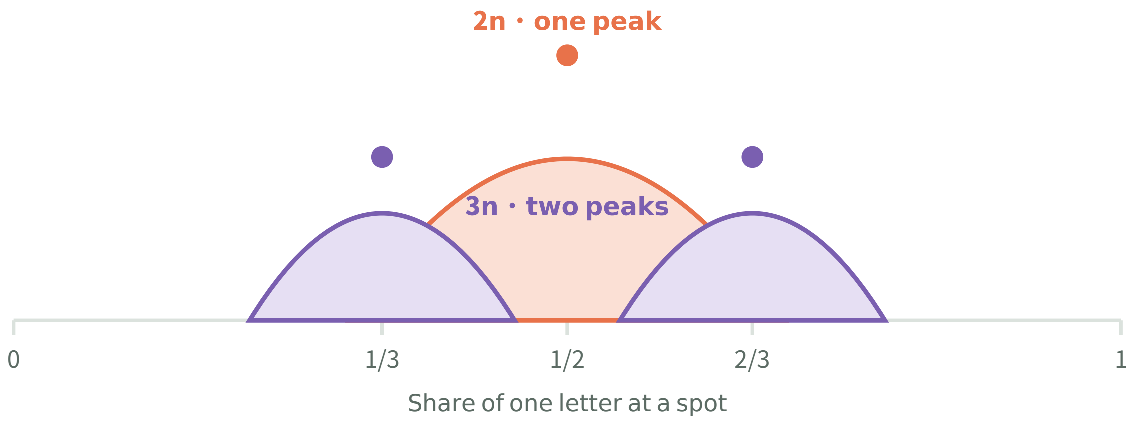
5. **Cross-check with many methods**
   See whether independent evidence agrees.

6. **Integrated call**
   Combine evidence — conflicts are shown, not hidden.

QUESTION 1 · KEY IDEA

Look at the mix ratio of two letters

When a spot has two letters — 2 strings split **50:50 (½)**, 3 strings split into **1/3 · 2/3**.

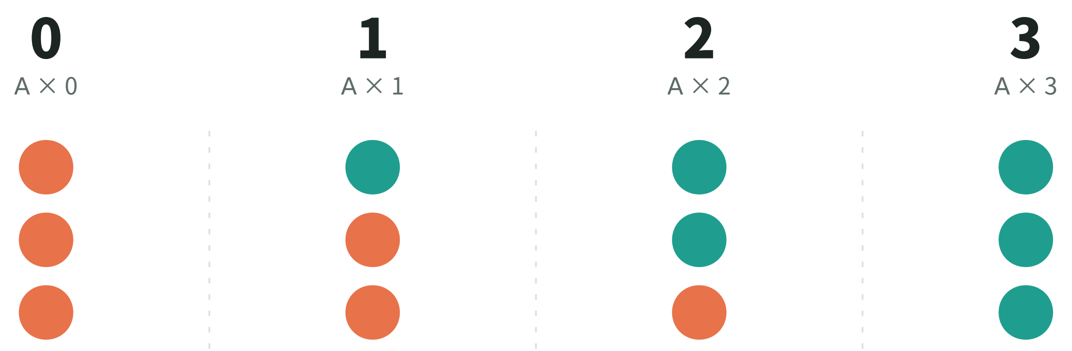


2 strings → an even split; 3 strings → a 1:2 split, so two peaks appear.

QUESTION 2 · KEY IDEA

Count how many strings are parent A

Using **parent-only markers (diagnostic SNPs)**, count whether A is 0, 1, 2 or 3 of the 3 strings.



Markers are picked **only from pure parents**, never from the triploid itself. (No copying its own answer.)

We don't trust just one clue

Check whether independent methods reach the same answer.

Chromosome painting

Color A/B tracts along each chromosome (HMM local ancestry).

D-statistic

Weigh which parent's letters are shared more.

k-mer

Cross-check with short letter-chunks, no mapping needed.

COX1

Confirm the maternal line via mitochondria.

Reciprocal mapping

Swap A/B references to check bias.

Diploid baseline

Calibrate against a confirmed 2n control.

Finally · an honest scorecard

Accept only when methods **agree**; if they clash, **show it as is**.

● **Supported**

Enough independent evidence agrees.

● **Needs checking**

If calibration fails or evidence is thin, mark `missing`.

● **Discordant**

If pedigree and COX1 differ, keep it as `discordant`.

This tool does not pin down an exact backcross generation — that needs parent genotypes. Instead it shows **whether the pattern fits a simple cross**.